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Association between environmental and headphone noise and heart rate variability: observations from Apple Hearing Study cohort

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BACKGROUND: Environmental noise is linked to adverse cardiovascular effects. However, few studies have examined people's heart rate variability (HRV) changes related to noise sources, despite the elevated cardiovascular risk associated with reduced HRV.
OBJECTIVES: We investigated HRV changes following short- and long-term lagged environmental and headphone sound exposure, assessed using the standard deviation of normal-to-normal intervals (SDNN). We also examined effect modification by individual characteristics.
METHODS: Eligible participants were a subset of consented adults from the Apple Hearing Study, a nationwide prospective cohort. Analyses were constructed into four analytical cohorts defined by sound source (environmental vs. headphone) and exposure window (short-term: 20 min vs. long-term: 160 min). In each cohort, twenty participants were randomly sampled from each unique age-sex-ethnicity group. Smartwatches and smartphones measured sound levels and 1-min SDNN estimates. Distributed lag non-linear hierarchical Bayesian models estimated lag-specific and overall changes in first-stage analyses, adjusting for time of day and physical activity, with random effects to capture individual-specific variation. Second-stage linear regression models summarized the average change by age, sex, BMI, stress, self-rated hearing ability, and tinnitus.
RESULTS: The stratified sampling ranged from 980 to 1180 participants across analyses. An overall SDNN reduction of 6.8% (short-term) to 16.0% (long-term) was observed following a 10-decibel increase in environmental noise exposure, and 7.1% (short-term) to 7.4% (long-term) for headphone sound. Lag-specific prediction showed brief initial SDNN increases followed by sustained reductions. Greater SDNN reductions were seen in older adults, individuals with tinnitus, or individuals who reported excellent hearing (for environmental noise) or those who reported poor hearing (for headphone sound).
SIGNIFICANCE: Findings suggested noise-HRV relationships differed by exposure window, noise source, and individual characteristics, highlighting the importance of stratified risk assessment, especially in the elderly and individuals with hearing loss. However, generalizability may be limited by cohort characteristics and reliance on consumer-grade wearables.
IMPACT: The physiological stress associated with noise exposure can persist from minutes to hours, as reflected by decreasing heart rate variability associated with elevated noise exposure levels. Given the association between chronically low heart rate variability and higher risk of cardiovascular diseases, noise source-specific risk assessments and noise control efforts are warranted for future studies on the non-auditory health effects, particularly for vulnerable populations.

Keywords: Personal exposure; Noise pollution; Cumulative impacts; Health studies; Sensors

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INTRODUCTION

Epidemiological studies have shown associations between long-term environmental noise exposure (especially traffic noise) and specific forms of cardiovascular disease (CVD), including higher prevalence of hypertension and ischemic heart disease [1–4]. The general stress model [5, 6] conceptualizes the causal pathway as a chain from sound perception to annoyance, to physiological stress arousal and, over time, to altered cardiovascular risk profiles [7]. Controlled laboratory exposure studies, involving both animal models and human subjects, have demonstrated that noise exposure can lead to neuroendocrine system activation, including an imbalance in the autonomous nervous system (ANS) and stress-hormone release [6, 8], which is consistent with this causal conceptualization. However, questions remain regarding the real-time ANS response under varying real-life noise exposures measured by wearable sensors in real-world, non-experimental settings.

A healthy heart continuously adjusts its beat-to-beat rhythms in response to physical and psychological stress through ANS regulation [9, 10], resulting in variations in consecutive heartbeat

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time intervals. Such variations are captured by heart rate variability (HRV), a widely used non-invasive indicator of ANS activity, including the sympathetic (SNS) and parasympathetic nervous systems (PNS) [11–13]. This indicator can be quantified using time-domain indices such as standard deviation of normal-to-normal heartbeat intervals (SDNN) for overall variation, or frequency-domain indices such as high-frequency band (HF, reflecting PNS activities) and low-frequency band (LF, influenced by both PNS and SNS activity and blood pressure regulation) [14]. At the population level, most time- and frequency-domain HRV indices decline with age, and females generally show lower SDNN than males [15, 16]. Overweight, measured by body mass index (BMI), has also been associated with reduced HRV [17]. At the individual level, HRV follows a circadian pattern that peaks in the early morning and drops to its lowest in the late afternoon [15]. Reduced HRV is associated with higher stress levels in the short term [18] and elevated long-term cardiovascular risk for ischemic heart disease and all-cause mortality [11, 19, 20], making it a suitable biomarker for examining the acute physiological consequences of noise exposure.

Several field studies found that higher noise levels are associated with reduced HRV–a trend consistent with sympathetic activation–typically within a 15-min window [21–23]. Meanwhile, evidence from exercise physiology indicates that sympathetic activation can persist for 45 to 60 min following intensive physiological arousal [24]; therefore, it is worth examining how noise-related autonomic response changes beyond 15 min. Additionally, sensitivity to noise may vary under conditions such as age, hearing ability, tinnitus, and perception of noise; whether and how the noise-HRV relationship may vary by these factors has not been examined.

With these considerations, this study aims to address the following questions: (1) how the changes in SDNN vary over time (for the prior 120 mins of noise exposure), and (2) how the noise-SDNN association varies by individual characteristics. To address these questions, we used a distributed lag non-linear model with random slopes to capture both the temporal structure of noise effects and individual variation in the exposure-response relationship. Exposures, measured as equivalent sound levels (L_{EQ}), were examined separately by source and time frame, as detailed in the Methods section. Findings from this study contribute to the understanding of the temporal dynamics and individual heterogeneity in the autonomic response associated with real-world noise exposure.

METHODS

Study population

This analysis extracted observational data from consented adults enrolled in an ongoing nationwide prospective cohort from November 2019 to December 2024, the overall cohort described in Neitzel et al. [25]. All study protocols were reviewed and approved by the IRB (Advarra, Inc, approval number Pro00037864). To be eligible in this analysis, participants had (1) consent to enable and share their Headphone Audio Levels (HAL) or Environmental Sound Levels (ESL) and HRV measurement features through their iPhone; (2) contribute at least 20 consecutive SDNN measurements; (3) complete a baseline demographic (i.e., identified age, sex assigned at birth, ethnicity) and hearing status (self-reported hearing ability and tinnitus occurrence) survey within one week of enrollment.

We implemented a stratified random sampling strategy due to demographic representativeness and model computation feasibility. Participants were classified into unique age-sex-ethnicity groups (60 strata from six age groups × two sexes × five ethnicity categories listed in Table 1). Twenty participants were drawn from each stratum using simple random sampling without replacement. A fixed random seed was set to ensure reproducibility. Strata with fewer than 20 eligible participants were excluded; in practice, this affected predominantly older participants. The inclusion process is demonstrated in Supplementary File Fig. S1.

Study design

The primary data consisted of repeated measurements nested within participating individuals. Four exposure history tables were constructed with time-related variables: SDNN (in milliseconds, ms); sound exposure (equivalent sound levels, L_{EQ} in A-weighted decibels, dBA); physical activity (energy expenditure, in units of kilocalories per hour, kcal/h); and time of day. Each row represented one SDNN observation with columns of sound exposure history from the latest lagged interval to the earliest lagged interval and additional columns for covariates and subject ID. Additional individual-level variables—age, sex assigned at birth, body mass index (BMI), self-rated hearing, tinnitus, and perceived stress—were included for individual-level variation analysis.

HRV and time variables. Among HRV indices, SDNN was selected as the primary outcome for this study because it provides a global measure of autonomic regulation that captures the net changes from sympathovagal shifts without requiring assumptions about the relative contributions of individual autonomic branches, and it was directly available from the wearable device used in this study. SDNN was recorded through 1-min tachograms collected at 2- to 4-h intervals when the participant was still, a condition automatically evaluated by Apple Watch (Apple Inc., Cupertino, CA, USA) via accelerometers [26]. Prior validation studies reported that Apple Watch-measured SDNN was systematically lower than the reference value up to 8.31 ms [27, 28]. SDNN values were log-transformed to approximate a normal distribution for model fitting. To account for circadian variation in SDNN [29], measurement time of day was categorized into morning (reference level, 6am-noon), afternoon (noon-5 pm), evening (5pm-9pm), and night (9pm-6am).

Noise exposure. Long-term environmental noise exposure was integrated as 20-min A-weighted equivalent sound levels ($L_{EQ,20\,min}$, see Supplementary File Eq. S1) from Environmental Sound Levels, which measured ambient sound levels through daily activities using the Noise application (app) on Apple Watch (Apple Inc., Cupertino, CA, USA). A prior validation study reported that the Noise app measured 2.1 dB and 3.2 dB lower than reference values under controlled conditions [30]. This may result in a conservative noise estimate in this study, although other artifacts (e.g., microphone rubbing against fabrics, water in the watch without using the water lock feature) may result in higher measurements. For each SDNN observation, lagged exposures were matched to it up to 160 min prior to create an exposure history table. Short-term environmental noise exposure history table matched 1-min equivalent sound levels ($L_{EQ,1\,min}$) up to 20 min in the same way. Additionally, headphone sound exposure was integrated from Headphone Audio Levels, which measured sound levels while streaming videos or listening to music/podcasts through headphones using the Health app on iPhone (Apple Inc., Cupertino, CA, USA). The distinction between "environmental noise" and "headphone sound" is based on the assumption that the majority of ambient sound exposure (e.g., traffic, construction) is unwanted and thus constitutes the "noise" (aligns with the definition from the United States Environmental Protection Agency [31]), whereas headphone-delivered sounds are predominantly voluntary and self-selected. We acknowledge that in practice, there is a mix of wanted and unwanted sound in both categories. Nevertheless, this distinction is made to reflect the differential role of annoyance–a key component in the general stress model [5, 6]–where involuntary environmental noise may be more likely to trigger annoyance and subsequent stress responses than voluntary headphone use.

Activity covariate. To adjust for activity intensity level and heart rate, which are known factors to affect SDNN [23, 32], we calculated concurrent 5-min active energy expenditure and included it as a control variable in the main models. This was derived from Apple's calorimetry algorithms [26], which consider heart rate to inform activity intensity along with other variables such as age, sex, weight, and motion-related signals. The value was log-transformed to match the outcome's numeric scale in the models.

Individual-level variables. Participants provided demographic data and completed surveys on hearing and stress. Weight (in kilograms), height (in centimeters), self-rated hearing ability (five levels, i.e., excellent, very good, good, fair, poor), and tinnitus frequency (five levels, i.e., never, a few times a year/month/week/day) were collected at baseline. BMI was calculated as weight/height^2, and categorized into underweight, normal, overweight, and class 1 to 3 obesity based on the World Health Organization guidance

Table 1. Participant characteristics by noise type and exposure duration, summarized into counts and percentages.

Demographics	Environmental Noise				Headphone Sound			
	Short-term $N = 1180$		Long-term $N = 1180$		Short-term $N = 1080$		Long-term $N = 980$	
Age								
18 ≤ age < 25	200	16.9%	200	16.9%	180	16.7%	140	14.3%
25 ≤ age < 35	200	16.9%	200	16.9%	200	18.5%	200	20.4%
35 ≤ age < 45	200	16.9%	200	16.9%	200	18.5%	180	18.4%
45 ≤ age < 55	200	16.9%	200	16.9%	200	18.5%	180	18.4%
55 ≤ age < 65	200	16.9%	200	16.9%	180	16.7%	180	18.4%
age ≥ 65	180	15.3%	180	15.3%	120	11.1%	100	10.2%
Assigned Sex								
Female	580	49.2%	580	49.2%	480	44.4%	420	42.9%
Male	600	50.8%	600	50.8%	600	55.6%	560	57.1%
Ethnicity								
Non-Hispanic White alone	240	20.3%	240	20.3%	240	22.2%	240	24.5%
Non-Hispanic Asian alone	220	18.6%	220	18.6%	200	18.5%	120	12.2%
Non-Hispanic Black alone	240	20.3%	240	20.3%	200	18.5%	180	18.4%
Hispanic	240	20.3%	240	20.3%	220	20.4%	220	22.4%
Multiethnic	240	20.3%	240	20.3%	220	20.4%	220	22.4%
BMI Class								
Underweight	22	1.9%	26	2.2%	21	1.9%	20	2.0%
Healthy	369	31.3%	362	30.7%	335	31.0%	284	29.0%
Overweight	384	32.5%	393	33.3%	355	32.9%	315	32.1%
Class 1 Obesity	201	17.0%	201	17.0%	181	16.8%	163	16.6%
Class 2 Obesity	102	8.6%	96	8.1%	94	8.7%	92	9.4%
Class 3 Obesity	82	6.9%	83	7.0%	78	7.2%	89	9.1%
Missing	20	1.7%	19	1.6%	16	1.5%	17	1.7%
Stress Score								
0 ≤ score < 2	164	13.9%	174	14.7%	170	15.7%	144	14.7%
2 ≤ score < 4	220	18.6%	224	19.0%	186	17.2%	163	16.6%
4 ≤ score < 6	228	19.3%	225	19.1%	204	18.9%	194	19.8%
6 ≤ score < 8	205	17.4%	206	17.5%	201	18.6%	178	18.2%
8 ≤ score < 10	181	15.3%	176	14.9%	159	14.7%	141	14.4%
10 ≤ score < 12	102	8.6%	99	8.4%	88	8.1%	90	9.2%
score ≥ 12	71	6.0%	68	5.8%	65	6.0%	66	6.7%
Missing	9	0.8%	8	0.7%	7	0.6%	4	0.4%
Hearing Ability								
Excellent	225	19.1%	225	19.1%	204	18.9%	177	18.1%
Very good	395	33.5%	402	34.1%	373	34.5%	333	34.0%
Good	346	29.3%	344	29.2%	308	28.5%	282	28.8%
Fair	169	14.3%	163	13.8%	150	13.9%	141	14.4%
Poor	45	3.8%	46	3.9%	45	4.2%	47	4.8%
Tinnitus Occurrence								
Never	276	23.4%	276	23.4%	261	24.2%	230	23.5%
A few times a year	431	36.5%	423	35.8%	388	35.9%	325	33.2%
A few times a month	172	14.6%	179	15.2%	157	14.5%	170	17.3%
A few times a week	111	9.4%	111	9.4%	119	11.0%	103	10.5%
A few times a day	190	16.1%	191	16.2%	155	14.4%	152	15.5%

[33]. Four weeks after enrollment, stress was self-rated using a 4-item Perceived Stress Scale (PSS-4) scaled from 0 (lowest stress) to 16 (highest stress) [34, 35]. Confirmatory factor analysis supported a two-factor structure for stress (helplessness and coping, each scale from 0 to 8), consistent with Schäfer et al. [36] and Ingram et al. [37]. Both one- and two-factor models were used in subsequent analyses for comparison; however, only one-factor model results were presented here for simplicity—two-factor model results were similar.

Statistical analysis

We describe our methodology briefly below; additional details are provided in the Supplemental File. The same modeling process was repeated for four analytical cohorts. All models used identical specifications and procedures.

All analyses were conducted using R (v4.5.0; R Foundation for Statistical Computing, Vienna, Austria) [38]. Distributed lag models were implemented using *dlnm* (v2.4.7) package [39]. Hierarchical Bayesian models were implemented using the *brms* (v 2.22.0) package [40]. Marginal means were implemented using the *emmeans* (v1.11.0) package [41].

Exposure-lag-response models

For each exposure history table, we applied a hierarchical Bayesian model with distributed lag non-linear terms [42–44] to estimate associations with lagged sound exposures. We assumed that the noise-SDNN relationship was time-varying due to instant regulation activity in the autonomic nervous system in response to previous exposures or activities [23], so a distributed lag model was more appropriate than a simple linear regression, given that previous events have a carryover effect on how the current event will change the outcome. The lagged noise exposure was constructed as the "*crossbasis*" function from *dlnm* (Eq. 7 in Gasparrini et al. [45]). Models also included the concurrent activity covariate and time of day as control variables.

The hierarchical structure of our models accounted for repeated measures within individuals and included random intercepts and slopes. This design allows us to capture between-subject variation in baseline SDNN and noise-SDNN relationships and to account for time-invariant confounders that were not directly measured (e.g., smoking, alcohol consumption, and pre-existing conditions). Residual confounding from acute time-varying behaviors such as episodic alcohol, caffeine intake, or sleep status may bias estimates in either direction, which remains a limitation. Intra-class correlation coefficients (ICC) showed 33 to 40% of model variance due to individual baseline differences, suggesting moderate-to-good clustering and supporting the use of hierarchical modeling. We used 2 degrees of freedom for the natural cubic spline basis, chosen to capture the essential non-linear structure with the fewest parameters and to avoid overfitting. A description of model specifications and prior settings is included in the Supplementary File.

The posterior baseline log-SDNN estimates of the study population for morning, afternoon, evening, and night were calculated at a reference exposure level of 40 dBA–representative of a quiet environment such as in a library–and at the median level of observed log-activity level, then transformed back to milliseconds and summarized as the median with a 95% credible interval.

Population- and individual-level posterior effect estimates at multiple exposure levels (i.e., L_{EQ} from 30 to 80 dBA by 1 dB increment) were computed using the *crosspred* function [39] from *dlnm* with posterior sample fixed effect coefficients and the sum of fixed and random effect coefficients for sound, respectively. The estimates were presented as the posterior sample median of the percent changes (%) in SDNN (r_i) at each lag at a specific exposure level relative to the reference SDNNs at 40 dBA. The overall change was the sum of all lag-specific changes over the entire lagged period, calculated at each specific exposure level. The predicted population-level overall changes and lag-specific changes were used for noise-SDNN visualization, while individual-level overall changes were used for the next step analysis.

Individual-level variations

The second stage analysis calculated the marginal mean and 95% confident interval by age groups, sex, BMI category, stress score, hearing status and tinnitus frequency using the simple linear regression models (Eq. 1) conditional on five noise levels (i.e., L_{EQ} at 30, 50, 60, 70, and 80 dBA), respectively:

$$r_i = \alpha_0 + \alpha_1 \text{Age}_i + \alpha_2 \text{Sex}_i + \alpha_3 \text{BMI}_i + \alpha_4 \text{Stress}_i + \alpha_5 \text{Hearing}_i + \alpha_6 \text{Tinnitus}_i + \epsilon_i \quad (1)$$

where r_i was the percent changes (%) in SDNN in the posterior prediction from Bayesian models (Eq. S2). We compared the estimated percent changes (%) at different levels of a single variable when exposed to same *LEQ* and holding other variables constant. For model simplicity, the uncertainty was not propagated from the Bayesian models (Eq. S2) to the second-stage model (Eq. 1); the impact of which is discussed later. Less than 2% of participants have missing BMI or stress scores, and the missingness appears to be at random; therefore, the regression models were conducted on complete cases.

Ethical approval

All analyses using data from the Apple Hearing Study were performed in accordance with relevant guidelines and recommendations. The Apple Hearing Study was approved by Advarra's Institutional Review Board (IRB; Approval Number: Pro00037864). All participants provided informed consent electronically prior to enrollment in the Apple Hearing Study.

RESULTS

Descriptive summaries

Four analytical cohorts were constructed based on sound source (environmental vs. headphone) and exposure window (short-term vs. long-term). For environmental noise analyses, 59 unique demographic combinations yielded a stratified sample of 1180 participants each (Table 1) from 49,983 (short-term) and 47,339 (long-term) eligible participants (Supplementary File Table S1). The headphone sound datasets included 54 and 49 strata for short- and long-term headphone models, respectively, resulting in a sample of 1080 from 19,006 (short-term) and 980 from 10,726 (long-term) eligible participants. Excluded strata were kept for sensitivity analysis. The sampling pool for headphone data was less than half the size of that of the environmental data, partially due to limited headphone use over a day. Despite differences in sample size, the four analytical cohorts had similar distributions of self-reported stress, hearing ability, and tinnitus occurrence.

For time-related variables (Table 2), median 1-min and 20-min average headphone sound levels (66.4 and 68.4 dBA) were higher

Table 2. Descriptive summary for three time-related variables and the number of observations in each analytical cohort.

Variables	Environmental Noise				Headphone Sound			
	Short-term $N = 361,629$		Long-term $N = 293,144$		Short-term $N = 160,389$		Long-term $N = 162,913$	
LEQ (dBA)	52.3 (43.2, 61.2)		62.0 (54.2, 68.4)		66.4 (58.3, 73.9)		68.4 (60.8, 75.0)	
SDNN (ms)	27.4 (17.0, 46.8)		27.3 (16.8, 47.1)		29.8 (19.8, 45.3)		29.7 (19.7, 45.8)	
Activity (kcal/h)	27.7 (19.3, 41.3)		28.0 (19.5, 41.8)		29.4 (19.8, 46.2)		31.5 (21.3, 49.4)	
Time of Day								
Morning	91,555	(25.3%)	82,369	(28.1%)	39,094	(24.4%)	35,488	(21.8%)
Afternoon	89,306	(24.7%)	74,878	(25.5%)	44,720	(27.9%)	36,676	(22.5%)
Evening	75,946	(21.0%)	64,577	(22.0%)	29,604	(18.5%)	34,059	(20.9%)
Night	104,822	(29.0%)	71,320	(24.3%)	46,961	(29.3%)	56,690	(34.8%)

Variables are summarized as median (1st quantile, 3rd quantile) or n (%).

than those of environmental noise levels (52.3 and 62.0 dBA). Median 1-min SDNN was 27.42 and 27.26 ms in environmental noise cohorts and 29.80 and 29.67 ms in headphone cohorts, respectively. The 5-min active energy expenditure followed the same pattern: 27.72 and 28.01 kcal/h in environmental cohorts versus 29.41 and 31.46 kcal/h in headphone cohorts.

Exposure-lag-response trends at the population level

The predicted morning SDNN values in median with 95% credible interval at the reference condition (40 dBA, median log-activity), were 37.5 [28.7, 48.8] ms, 35.9 [27.0, 47.7] ms, 39.7 [29.3, 47.4] ms, and 32.7 [25.5, 41.8] ms for the short-term environmental noise, long-term environmental noise, short-term headphone sound, and long-term headphone sound exposure cohorts (Supplementary File, Table S2).

Compared to the simple moving average models commonly used in other literature for time series analysis, the distributed lag model does not assume the effect from a time further away contributes equally to the response as more recent exposures [45, 46] and allows time-varying exposure-response relationships to be plotted on a bi-dimensional surface (Supplementary File Fig. S2). Figure 1 shows the SDNN percent changes *predicted* at each sound level and time lag relative to the baseline level at 40 dBA. With environmental L_{EQ} elevated from 40 dBA to 50 dBA, for example, we observed median 0.2–0.7% increases in SDNN from each 1-min lag during the initial 6-min window. No meaningful changes were observed around 7–8 min while reductions (0.2–1.4%) emerged from 8 to 20 min (Fig. 1a), resulting in a net overall reduction of 6.8 [5.1, 8.4] % at 50 dBA (Fig. 2a, red line). In comparison, the prediction showed consistent reductions (1.4–2.9%) in SDNN across the 160-min lag window when environmental noise level was elevated to 50 dBA, with smaller contribution from earlier exposures (Fig. 1b), and a net overall reduction of 16.0 [13.5, 18.3] % (Fig. 2a, blue line).

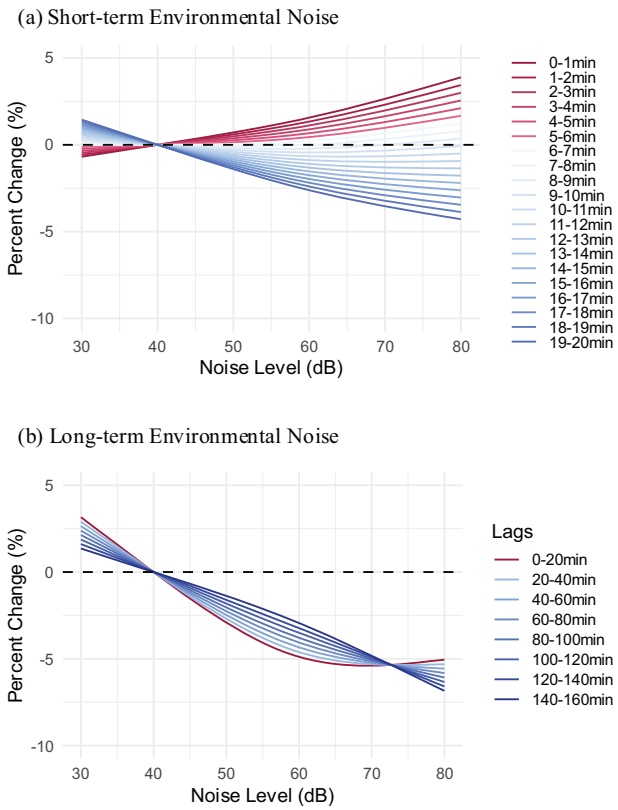


Fig. 1 Predicted median lag-specific percent changes (%) in SDNN from environmental noise exposures. Estimates reflect percent changes in SDNN relative to predicted SDNN at 40 dBA. **a** Estimates for short-term environmental noise exposure. **b** Estimates for long-term environmental noise exposure.

The predicted SDNN changes in the headphone sound models followed a slightly different temporal pattern. There was a 0.1–0.2% increase during the first 3 min, followed by no changes during the third and fourth minutes and an eventual 0.1–0.9% reduction after 5 min (Fig. 3a), with a net overall reduction of 7.1 [5.0, 9.0] % at 50 dBA (Fig. 2b, red line). The prediction from the long-term headphone sound model showed initial SDNN reductions by 1.4 [0.4, 2.5] % but the changes flattened toward zero (–0.5 [–1.6, 0.5] %) in the last 20-min lag (Fig. 3b) and the net overall changes were –7.4 [–9.9, –4.9] % at 50 dBA (Fig. 2b, blue line).

In sensitivity analyses restricted to excluded strata (Supplementary File, Table S5), predicted morning SDNN at the reference condition was slightly lower (median 29.58 to 36.24 ms) than in the included population, as expected given their older age. The noise-SDNN pattern—a brief increase followed by a sustained reduction—was consistent with that in the included population, with no meaningful differences in the magnitude of change.

Looking at the predicted overall changes across all exposure levels (Fig. 2), greater reductions were observed with elevated L_{EQ}, except that short-term environmental noise exposure showed a slightly U-shaped curve, with the largest reduction of 10.4 [8.1, 12.7] % around 60 dBA. This turning point lies close to the 65 dBA cut-off value reported in Kraus et al. 2013, where they observed different noise-SDNN associations. The attenuated reduction at the high short-term levels may relate to fewer data points at the upper end of the exposure range. The overall reduction was 27.4 [24.1, 30.4] % at 60 dBA for long-term exposures, more than two times that of short-term exposures. For headphone sound models, short- and long-term overall changes were similar, linked to 14.3 [11.3, 17.0] % and 14.6 [11.0, 18.1] % predicted reduction at 60 dBA, respectively.

The HMC diagnostics [47] indicated good convergence and chain mixing for all four Bayesian models: all $\hat{R}$< 1.01 (convergence indicator), effective sample sizes were greater than 10% of post-warm-up iterations, and visual inspection of trace plots showed no concerning patterns among the chains. Q-Q plots showed slight deviation from normality at each end and were not sensitive to alternative priors. Conditional R^2 values measure the proportion of variance explained by both noise exposure levels and variation across individuals combined, and were 0.609 (95% CI: [0.608, 0.611]), 0.621 [0.619, 0.622], 0.431 [0.429, 0.435], and 0.441 [0.438, 0.443] for short-term environmental, long-term environmental, short-term headphone, and long-term headphone models, respectively. Marginal R^2 values measure the proportion of variance explained by noise exposure only and were 0.007 (95% CI: [0.007, 0.008]), 0.015 [0.013, 0.017], 0.018 [0.015, 0.019], and 0.013 [0.011, 0.015] for the four models, respectively. The small marginal R^2 indicated that the noise exposure alone explains a relatively small proportion of the variance in SDNN. Intra-class correlation coefficient (ICC) further measures the variation due to individual differences. ICC for random intercept ranged from 0.33 to 0.40 consistently (Table 3), showing moderate-to-good clustering by individuals and supporting the inclusion of an individual-specific intercept to avoid underestimating uncertainty. The total ICC (i.e., including both random intercept and random slope) significantly increased in long-term exposure-response models, justifying the inclusion of individual-specific sound-HRV responses instead of assuming universal effects across individuals.

Overall responses variation at the individual level

Individual-specific overall SDNN changes in median, derived from posterior distributions with both fixed and random effects for sound, were summarized into mean and 95% CI across participant characteristics (Supplementary File, Tables S3 and S4). We

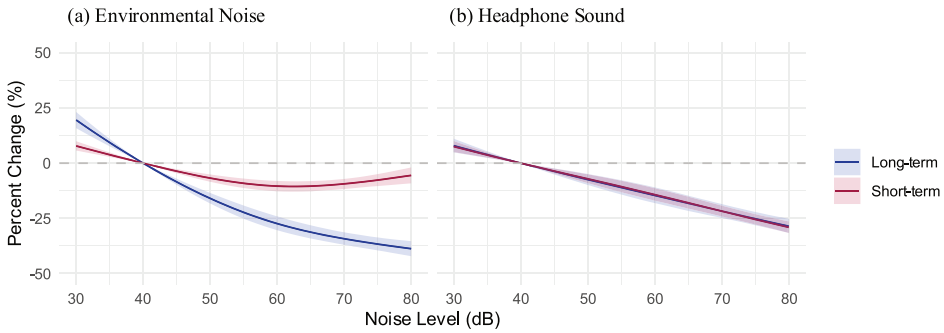


Fig. 2 Predicted median overall percent change (%) in SDNN across noise levels from short- and long-term exposure groups, with shaded area representing 95% credible interval. Estimates reflect percent changes in SDNN relative to predicted SDNN at 40 dBA. **a** Estimates for environmental noise exposure. **b** Estimates for headphone sound exposure.

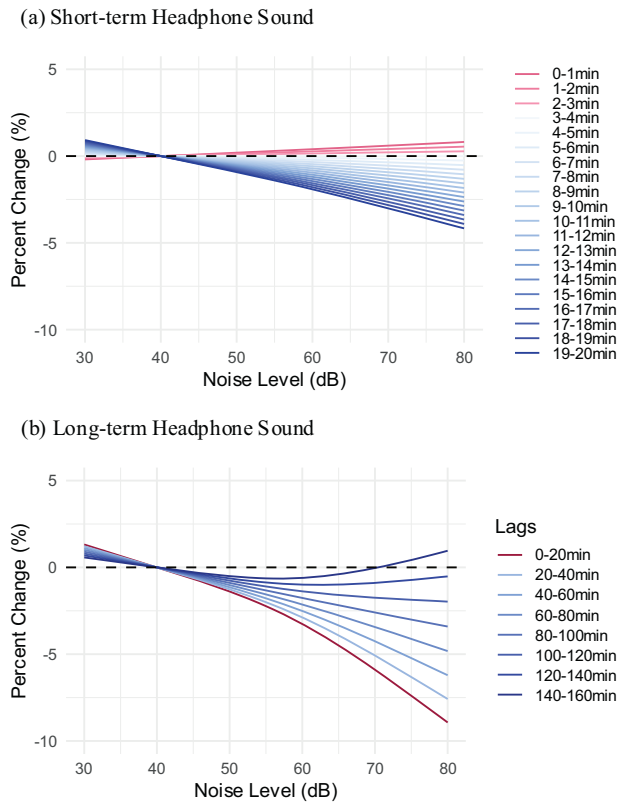


Fig. 3 Predicted median lag-specific percent changes (%) in SDNN from headphone sound exposures. Estimates reflect percent changes in SDNN relative to predicted SDNN at 40 dBA. **a** Estimates for short-term headphone sound exposure. **b** Estimates for long-term headphone sound exposure.

compared the mean across all levels for each variable within each exposure level. Because the uncertainty of the relative change estimate was not inherited into the individual-level model, the 95% Confidence Interval (95% CI) width below may be underestimated, and therefore, the comparisons did not include significance testing.

Responses to long-term environmental exposure

Older adults showed greater average changes in response to environmental noise exposure compared to younger participants. For example, at 60 dBA, the estimated mean percent change was −20.9% (95% CI: [−22.6, −19.3] %) in the 18–25 age group, and −31.3 [−33.1, −29.4] % in participants aged 65 and older. At 30 dBA, the percent change in the 65+ group (26.7 [24.3, 29.1] %) doubled that of the youngest group (13.8 [11.6, 15.9] %). Both predicted trends showed older individuals having more drastic changes in both directions. Compared to the healthy weight group, higher BMI was also associated with larger percent changes in both directions. For instance, Class 3 obesity had a change by 23.9 [21.0, 26.8] % at 30 dBA and −29.6 [−31.8, −27.3] % at 60 dBA, the absolute values were larger than those with healthy weight (18.3 [16.7, 19.9] % at 30 dBA and −24.9 [−26.1, −23.6] % at 60 dBA). Similar differences were observed in Class 1 and Class 2 obesity between 30 dBA and 70 dBA. The association between SDNN and noise was not different by sex or stress level. Although, compared to a moderate stress level (scored between 6 to 8), higher stress (score between 8 to 12) showed a slightly larger relative reduction given the same exposure level changes.

The predicted SDNN changes differed across levels of self-rated hearing ability at exposures of 70 dBA and above. At 70 dBA, participants who rated their hearing as excellent (−35.4 [−37.4, −33.4] %) exhibited a 4.9% greater change than those who reported poor hearing (−30.5 [−34.3, −26.7] %). At 80 dBA, the estimated difference was even greater (13.1%).

Reporting tinnitus occurrence showed greater changes compared to participants who never had tinnitus. For example, at 60 dBA, participants whose tinnitus happened a few times a year (−28.1 [−29.5, −26.7] %), or a few times a day (−28.3 [−30.0, −26.7] %) had greater SDNN reduction than participants who reported never having tinnitus (−25.6 [−27.2, −24.0] %).

Short-term environmental noise exposure showed qualitatively similar within-variable trends across age, stress, hearing, and tinnitus groups, but with smaller effect magnitudes. In general, the variables included in the models explained a small fraction (less than 20%) of between-subject variation in both models, and the short-term models explained even less variation than the long-term models.

Responses to long-term headphone exposure

The predicted SDNN percent changes to long-term headphone sound exposures across age groups and BMI categories mirrored the patterns observed in environmental noise models, except that the changes were near-zero in younger adults (<25 years) and at lower exposure levels (<60 dBA). No trend was observed between the sexes.

Because headphone sound exposure is more likely to be voluntary, desirable, or beneficial (e.g., for stress relief) [48] than environmental noise, within-variable trends may differ between these sources. Perceived high stress showed greater percent reductions in response to headphone sound. For instance, at 60 dBA, those scored most stressed (PSS-4 scores at 12 and above) had an average percent change of −18.8 [−21.7, −16.0] %, compared to −13.1 [−15.2, −10.9] % for those less stressed (scores lower than 2). Additionally, participants with poorer self-rated hearing stood out as having more changes (−25.6 [−28.5, −22.7] %

Table 3. Intra-class correlation coefficients (ICCs) considering random intercept only ($ICC_{intercept}$) versus considering both random intercept and random slope (ICC_{total}) across analytical groups, with point estimates and 95% Confidence Intervals.

	Environmental Noise		Headphone Sound	
Variables	**Short-term $N = 361,629$**	**Long-term $N = 293,144$**	**Short-term $N = 160,389$**	**Long-term $N = 162,913$**
$ICC_{intercept}$	0.40 (0.38, 0.43)	0.38 (0.35, 0.41)	0.37 (0.35, 0.40)	0.33 (0.30, 0.36)
ICC_{total}	0.41 (0.39, 0.44)	0.50 (0.47, 0.52)	0.40 (0.38, 0.43)	0.42 (0.38, 0.46)

at 70 dBA and −32.5 [−37.4, −27.7] % at 80 dBA) than excellent hearing participants (−22.3 [−24.0, −20.6] % at 70 dBA and −26.9 [−29.7, −24.1] % at 80 dBA). This trend is the opposite of that of environmental noise exposure. Lastly, at 50 to 70 dBA, participants who reported having tinnitus a few times a year, a month, and a week had a larger reduction than participants who never had tinnitus or had tinnitus a few times a day.

Short-term headphone exposures showed fewer clear subgroup differences. Both short- and long-term models explained up to 23% of between-subject variability in SDNN changes, with short-term models explaining slightly less.

DISCUSSION

This analysis examined the percent SDNN changes (%) following two types of sound exposures over short- and long-term lagged periods, using real-world data from consumer wearables. Predicted net SDNN reductions were observed at elevated exposure levels for both environmental noise and headphone sound. Lag-specific predictions showed a brief period of SDNN increase in the first 3–6 min, followed by sustained reduction. Our hierarchical Bayesian models demonstrated heterogeneity across individuals by characteristics such as common demographics and hearing status.

The negative trend in predicted SDNN changes with increased noise exposure is consistent with findings by Walker et al. [18], who reported reduced 5-min SDNN during 40-min environmental noise exposure. Prior studies also found immediate SDNN increases followed by lagged decreases within 15 min with louder environmental noise [21–23], a trend consistent with current results (Figs. 1b and 3b). They attributed the observed early increase to changes in the very-low frequency band (VLF) of HRV [23] or sudden heart rate shifts [21]. In our case, the 1-min window for SDNN is too short to adequately capture VLF cycles, which require at least 5-min [49], but sudden heart rate shifts are likely to drive early SDNN increase. However, our observational design cannot isolate the mechanism, and this hypothesis would be best addressed through laboratory-controlled experiments.

A key contribution of this study is our observation of heterogeneous exposure-response relationships across lags and individual-level characteristics, especially for long-term exposure (greater total ICC gain when including random slope compared to short-term, Table 3). Older adults showed larger SDNN reductions compared to younger adults with the same L_{eq}, while controlling for hearing ability and tinnitus. This suggests greater autonomic disruption in older adults under noise exposure at equivalent levels. Sensorineural aging and hyperacusis, characterized by reduced sound tolerance and frequently a comorbidity with stress-related conditions, are more prevalent in older adults [50, 51], potentially increasing susceptibility to noise-related cardiovascular stress.

The direction of the HRV changes differed between environmental noise and headphone sound contexts. Among those exposed to environmental noise $\geq$70 dBA, the greatest SDNN reduction was observed in participants with excellent hearing (long-term: 35.4–40.2%, short-term: 6.4–11.3%). For headphone sound, the largest reductions were observed among participants with poor hearing (long-term: 25.6–32.5%, short-term: 22.6–29.2%). This opposite trend may reflect differences in listening context. Headphone sound, often associated with active listening (e.g., music, podcasts), may impose greater cognitive demand on individuals with hearing difficulties, increasing listening effort, increasing stress, and reducing HRV [51, 52]. In contrast, individuals with excellent hearing, or possibly hyperacusis, may respond more strongly to unwanted ambient noise, being more easily bothered or distracted by it; however, this hypothesis requires further investigation. Notably, the study did not account for sound characteristics or hearing protection use due to limited information. However, less than 10% of participants reported wearing hearing protection, and the duration of wearing was limited. The SDNN changes following the low- and high-frequency noise were not meaningfully different [18]. We acknowledge their potential confounding in the noise-SDNN relationship but expect limited impact in this study. Future studies adding ecological momentary assessment of the two variables would help improve exposure characterization.

Beyond the caveats noted above, several broader limitations apply. First, the stratified sample is composed of current Apple product users, and therefore the results may not be generalizable to the US population [25]. Due to small stratum sizes, some subgroups (e.g., Asian females aged 65 +) were excluded, which may limit generalizability to these groups; however, their exclusion did not materially change the observed associations. Second, regarding exposure and outcome measurement, the consumer-grade wearables were also not individually calibrated, and existing validation studies cover only a subset of Apple Watch models [28, 30]; development of a validation database for commonly used wearables would benefit future real-world exposure assessment research. In addition, HRV was limited to SDNN owing to data availability, and other indices may offer a richer characterization of autonomic nervous system responses to noise. Third, regarding modeling, we only partially accounted for heart rate changes in the model with the inclusion of active energy expenditure, while SDNN is inversely correlated with heart rate. Future work should consider simplifying the model to include heart rate and other lifestyle factors to examine how heart rate modifies the association between noise and SDNN. Lastly, to include uncertainties from first-stage estimation, a meta-regression approach will be explored to allow hypothesis testing for effect modification.

CONCLUSION

To our knowledge, it is among the first to characterize short- and long-term sound exposure and HRV at scale using real-world consumer-wearable data, capturing free-living exposures that laboratory studies of brief, controlled noise cannot. The distributed lag non-linear modeling further allowed us to resolve heterogeneous, time-varying exposure–response relationships. We observed overall SDNN reductions of 6.8-16.0% for each 10 dB increase in environmental noise exposure, and 7.1-7.4% for a 10 dB increase in headphone sound. Greater SDNN reduction was seen in older adults and individuals with tinnitus. Participants who reported excellent hearing showed greater reduction from

loud environmental noise exposure, but for headphone sound, it was the poor hearing participants who showed greater reduction. The findings highlight the role of noise context (environmental versus headphone source) in the noise-SDNN relationships and the importance of stratified risk assessment, especially in older adults and individuals with hearing loss. However, generalizability may be limited by cohort characteristics and uncalibrated consumer-grade wearables. Future work to include heart rate directly into the model or replace SDNN may provide additional insight. Additionally, it would be of interest to examine the chronic association using daily average SDNN and noise exposures, with careful consideration of various important and related lifestyle factors.

DATA AVAILABILITY

Aggregated, de-identified data that support the findings of this study may be available upon request from the corresponding author (RLN). All requests will be reviewed and considered in accordance with policies designed to protect participant confidentiality and to remain consistent with the study protocol and the consent provided by participants.

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AUTHOR CONTRIBUTIONS

XZ: conceptualization, data curation, formal analysis, investigation, original draft preparation, review and editing; SKP: resources, supervision, review and editing; LMS: project administration, resources, review and editing; RLN: supervision, conceptualization, validation, funding acquisition, review and editing.

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COMPETING INTERESTS

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